

Control Flow Syntax Analyzer & Parser

1. Project Overview

The Control Flow Syntax Analyzer & Parser is a compiler front-end system developed to analyze and validate the syntax of structured programming constructs. The project focuses specifically on control flow statements, including conditional (if, if-else) and iterative (for) constructs.

The system uses a Context-Free Grammar (CFG) to describe valid program structures. It ensures that input source code strictly follows syntactic rules. The implementation is carried out using ANTLR within a Java environment.

The final output includes syntax validation and an Abstract Syntax Tree (AST) representing the structure of the code.

2. Objectives

2.1 Lexical Analysis

The system breaks source code into tokens such as keywords, identifiers, operators, and delimiters.

2.2 Grammar Design (CFG Construction)

A CFG is designed to define valid syntax for control flow constructs, supporting nesting and recursion.

2.3 Syntax Analysis (Parsing)

The parser checks if the tokens follow grammar rules and detects syntax errors.

2.4 AST Construction

The system generates an AST representing hierarchy and execution flow.

3. System Design and Implementation

Tools used:

- ANTLR for parser generation

- Java for implementation

Workflow:

Input → Tokenization → Parsing → Validation → AST generation

4. Key Challenges

4.1 Nested Structures

Handling loops and conditions inside each other.

4.2 Dangling Else Problem

Ensuring correct association of else statements.

4.3 Scope Boundaries

Handling single-line and block statements correctly.

5. Testing and Validation

Test cases include:

- Simple if statements
- if-else structures
- for loops
- Nested and deeply nested structures
- Empty blocks